Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: John Brasil Dairy #2

Physical address of dairy:

Number and Street	City	County	Zip Code
9018 W Linwood AVE	Turlock	Stanislaus	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X058-X170-X018-XXXX

B. OPERATORS

Brasil, John

Operator name: Brasil, John

Telephone no.: (209) 632-7867 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
2613 S Mitchell RD	Turlock	CA	95380

This operator is responsible for paying permit fees.

C. OWNERS

Waterson, Arlon

Legal owner name: Waterson, Arlon

Telephone no.: (209) 652-3897 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
8150 Lander AVE	Hilmar	CA	95324

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	752	0	0	0	0	0
Maximum number	752	72	82	92	82	0
Average number	752	72	82	92	82	0
Avg live weight (lbs)	1,200	1,250	750	500		

Predominant milk cow breed: Jersey

Average milk production: 60 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 21,280.93 *tons per reporting period*

Total nitrogen from manure: 274,509.50 *lbs per reporting period*

After ammonia losses (30% loss applied): 192,156.65 *lbs per reporting period*

Total phosphorus from manure: 45,400.43 *lbs per reporting period*

Total potassium from manure: 139,074.30 *lbs per reporting period*

Total salt from manure: 370,635.60 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 7,290,000 *gallons*

Total nitrogen generated: 27,704.28 *lbs*

Total phosphorus generated: 7,011.30 *lbs*

Total potassium generated: 68,681.02 *lbs*

Total salt generated: 281,603.69 *lbs*

	7,290,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	7,290,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - DW #1	Ground water
Irrigation Water - Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Jersey North	8	8	2	process wastewater	X058-X170-X018-XXXX
Jersey South	12	12	2	process wastewater	X058-X170-X018-XXXX
Jersey West	40	40	2	process wastewater	X058-X170-X018-XXXX
Totals for areas that were used for application	60	60	6		
Totals for areas that were not used for application					
Land application area totals	60	60	6		

B. CROPS AND HARVESTS

Jersey North

Field name: Jersey North

11/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 8 Plant date: 11/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	135.55 *ton*	Dry-weight		83.7	20,500.00	4,300.00	41,700.00		14.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.94	113.24	23.75	230.34	778.84

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 8 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	263.90 *ton*	Dry-weight		72.0	11,500.00	2,500.00	14,000.00		4.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.99	212.44	46.18	258.62	868.23

Jersey South

Field name: Jersey South

11/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 12 Plant date: 11/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	199.74 *ton*	Dry-weight		84.5	20,200.00	4,800.00	40,700.00		13.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.65	104.23	24.77	210.01	706.91

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	395.55 *ton*	Dry-weight		72.3	11,600.00	2,500.00	13,100.00		4.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.96	211.83	45.65	239.22	785.23

Jersey West

Field name: Jersey West

11/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	678.77 *ton*	Dry-weight		84.4	20,500.00	4,600.00	37,000.00		13.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.97	108.54	24.35	195.89	725.33

Jersey West

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,328.55 *ton*	Dry-weight		71.3	13,200.00	2,600.00	13,200.00		4.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	33.21	251.65	49.57	251.65	934.17

NUTRIENT BUDGET

A. LAND APPLICATIONS

Jersey North - 11/19/2022: Oats, silage-soft dough

Field name: Jersey North

Crop: Oats, silage-soft dough — Plant date: 11/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.53	875,550.00 *gal*
Application event totals		0.00	0.00	0.00	36.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	74.17	18.69	188.60	781.09	160,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.13	841,875.00 *gal*
Application event totals		74.17	18.69	188.60	816.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	69.53	17.52	176.81	732.27	150,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.94	909,225.00 *gal*
Application event totals		69.53	17.52	176.81	770.21	

Jersey North - 05/30/2023: Corn, silage

Field name: Jersey North

Crop: Corn, silage — Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Jersey North - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.94	909,225.00 *gal*
Application event totals		0.00	0.00	0.00	37.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.56	1,043,925.00 *gal*
Application event totals		0.00	0.00	0.00	43.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	153.18	42.25	168.99	344.23	150,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.94	909,225.00 *gal*
Application event totals		153.18	42.25	168.99	382.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.56	1,043,925.00 *gal*
Application event totals		0.00	0.00	0.00	43.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	74.17	18.69	188.60	781.09	160,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.94	909,225.00 *gal*
Application event totals		74.17	18.69	188.60	819.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	54.80	1,313,325.00 *gal*
Application event totals		0.00	0.00	0.00	54.80	

Jersey North - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	60.26	15.19	153.24	634.64	130,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.75	976,575.00 *gal*
Application event totals		60.26	15.19	153.24	675.38	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	56.20	1,347,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.20	

Jersey South - 11/19/2022: Oats, silage-soft dough

Field name: Jersey South

Crop: Oats, silage-soft dough

Plant date: 11/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	39.34	1,414,350.00 *gal*
Application event totals		0.00	0.00	0.00	39.34	

03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	71.08	17.91	180.74	748.55	230,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.53	1,313,325.00 *gal*
Application event totals		71.08	17.91	180.74	785.08	

Jersey South - 11/19/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	67.99	17.14	172.88	716.00	220,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.53	1,313,325.00 *gal*
Application event totals		67.99	17.14	172.88	752.53	

Jersey South - 05/30/2023: Corn, silage

Field name: Jersey South

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.53	1,313,325.00 *gal*
Application event totals		0.00	0.00	0.00	36.53	

06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.28	1,448,025.00 *gal*
Application event totals		0.00	0.00	0.00	40.28	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	120.53	30.38	306.47	1,269.27	390,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	38.41	1,380,675.00 *gal*
Application event totals		120.53	30.38	306.47	1,307.68	

Jersey South - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.53	1,313,325.00 *gal*
Application event totals		0.00	0.00	0.00	36.53	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.71	23.37	235.75	976.37	300,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.28	1,448,025.00 *gal*
Application event totals		92.71	23.37	235.75	1,016.64	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.66	1,245,975.00 *gal*
Application event totals		0.00	0.00	0.00	34.66	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.26	19.47	196.46	813.64	250,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	33.72	1,212,300.00 *gal*
Application event totals		77.26	19.47	196.46	847.36	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	33.72	1,212,300.00 *gal*
Application event totals		0.00	0.00	0.00	33.72	

Jersey West - 11/19/2022: Oats, silage-soft dough

Field name: Jersey West

Crop: Oats, silage-soft dough

Plant date: 11/19/2022

Jersey West - 11/19/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.94	4,546,125.00 *gal*
Application event totals		0.00	0.00	0.00	37.94	

03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	76.02	19.16	193.31	800.62	820,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.94	4,546,125.00 *gal*
Application event totals		76.02	19.16	193.31	838.56	

04/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.61	17.29	174.45	722.51	740,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.69	4,276,725.00 *gal*
Application event totals		68.61	17.29	174.45	758.20	

Jersey West - 05/30/2023: Corn, silage

Field name: Jersey West

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.05	5,758,425.00 *gal*
Application event totals		0.00	0.00	0.00	48.05	

Jersey West - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.18	5,893,125.00 *gal*
Application event totals		0.00	0.00	0.00	49.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	111.26	28.04	282.90	1,171.64	1,200,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.05	5,758,425.00 *gal*
Application event totals		111.26	28.04	282.90	1,219.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.62	5,825,775.00 *gal*
Application event totals		0.00	0.00	0.00	48.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	112.18	28.27	285.25	1,181.40	1,210,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.81	5,489,025.00 *gal*
Application event totals		112.18	28.27	285.25	1,227.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.24	5,421,675.00 *gal*
Application event totals		0.00	0.00	0.00	45.24	

Jersey West - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	109.40	27.57	278.18	1,152.11	1,180,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.68	5,354,325.00 *gal*
Application event totals		109.40	27.57	278.18	1,196.79	

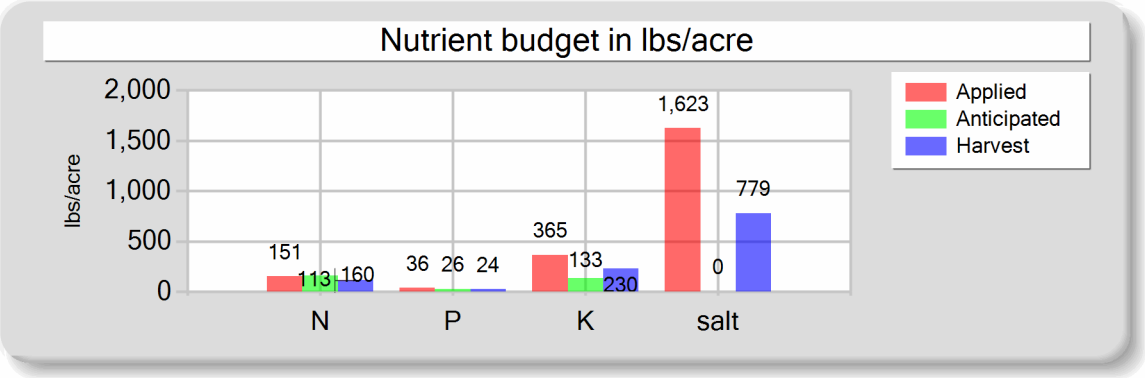
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.18	5,893,125.00 *gal*
Application event totals		0.00	0.00	0.00	49.18	

B. NUTRIENT BUDGET

Jersey North - 11/19/2022: Oats, silage-soft dough

Field name: Jersey North | Crop: Oats, silage-soft dough | Plant date: 11/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	143.71	36.22	365.41	1,513.37
Fresh water	0.00	0.00	0.00	109.60
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	150.71	36.22	365.41	1,622.96
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	113.24	23.75	230.34	778.84
Nutrient balance	37.47	12.47	135.07	844.13
Applied to removed ratio	1.33	1.52	1.59	2.08

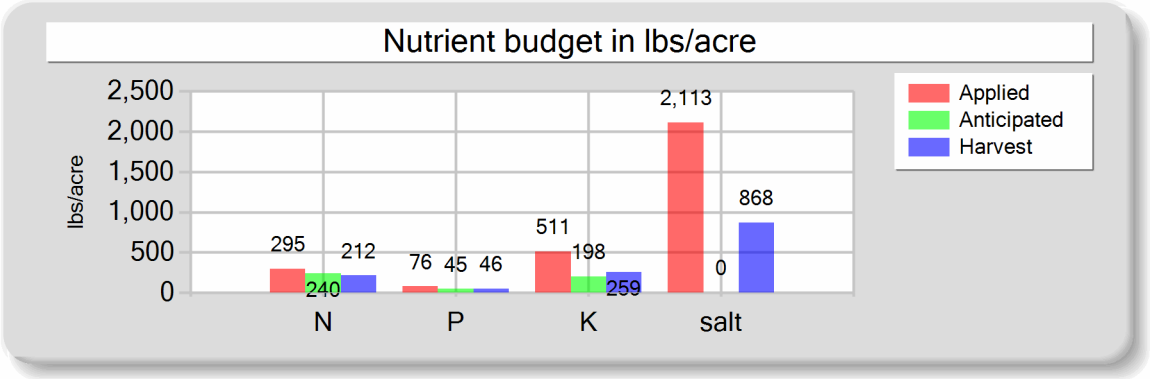
Fresh water applied	
2,626,650.00	*gallons*
96.73	*acre-inches*
12.09	*inches/acre*

Process wastewater applied	
310,000.00	*gallons*
11.42	*acre-inches*
1.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jersey North - 05/30/2023: Corn, silage

Field name: Jersey North | Crop: Corn, silage | Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	287.62	76.13	510.82	1,759.96
Fresh water	0.00	0.00	0.00	352.68
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	294.62	76.13	510.82	2,112.64
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	212.44	46.18	258.62	868.23
Nutrient balance	82.18	29.94	252.20	1,244.41
Applied to removed ratio	1.39	1.65	1.98	2.43

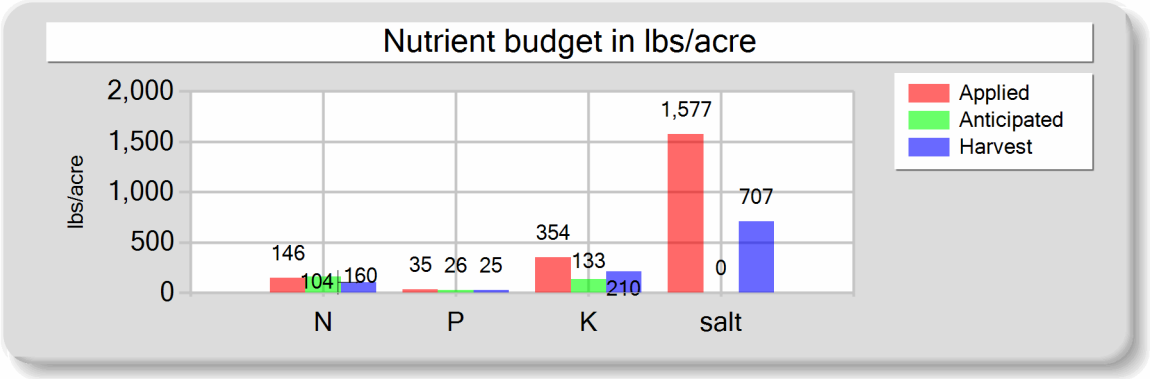
Fresh water applied	
8,452,425.00	*gallons*
311.27	*acre-inches*
38.91	*inches/acre*

Process wastewater applied	
440,000.00	*gallons*
16.20	*acre-inches*
2.03	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jersey South - 11/19/2022: Oats, silage-soft dough

Field name: Jersey South | Crop: Oats, silage-soft dough | Plant date: 11/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	139.07	35.05	353.62	1,464.55
Fresh water	0.00	0.00	0.00	112.41
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	146.07	35.05	353.62	1,576.95
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	104.23	24.77	210.01	706.91
Nutrient balance	41.84	10.28	143.61	870.04
Applied to removed ratio	1.40	1.42	1.68	2.23

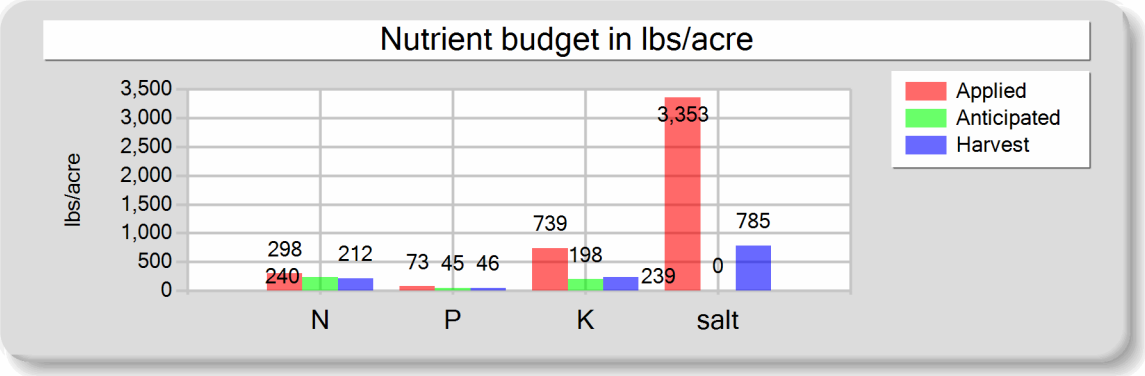
Fresh water applied	
4,041,000.00	*gallons*
148.82	*acre-inches*
12.40	*inches/acre*

Process wastewater applied	
450,000.00	*gallons*
16.57	*acre-inches*
1.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jersey South - 05/30/2023: Corn, silage

Field name: Jersey South | Crop: Corn, silage | Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	290.50	73.21	738.67	3,059.28
Fresh water	0.00	0.00	0.00	294.13
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.50	73.21	738.67	3,353.41
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	211.83	45.65	239.22	785.23
Nutrient balance	85.67	27.56	499.45	2,568.18
Applied to removed ratio	1.40	1.60	3.09	4.27

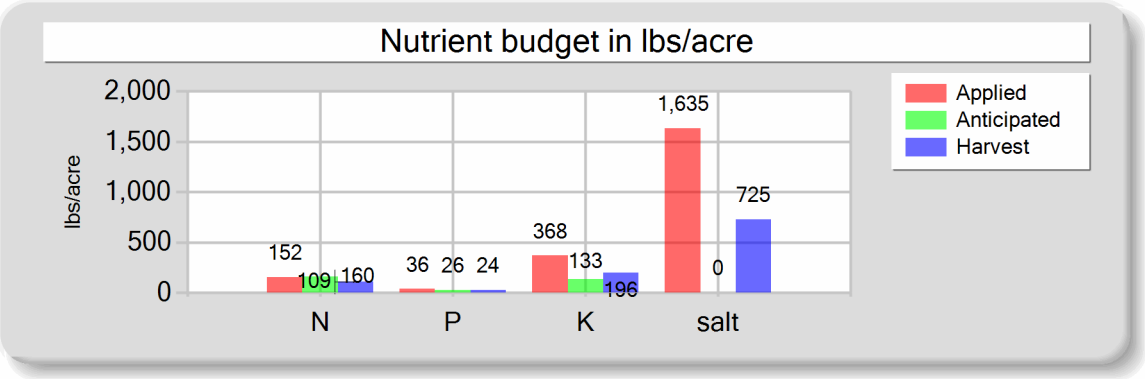
Fresh water applied	
10,573,950.00	*gallons*
389.40	*acre-inches*
32.45	*inches/acre*

Process wastewater applied	
940,000.00	*gallons*
34.62	*acre-inches*
2.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jersey West - 11/19/2022: Oats, silage-soft dough

Field name: Jersey West | Crop: Oats, silage-soft dough | Plant date: 11/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	144.63	36.45	367.76	1,523.13
Fresh water	0.00	0.00	0.00	111.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	151.63	36.45	367.76	1,634.69
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	108.54	24.35	195.89	725.33
Nutrient balance	43.10	12.10	171.87	909.36
Applied to removed ratio	1.40	1.50	1.88	2.25

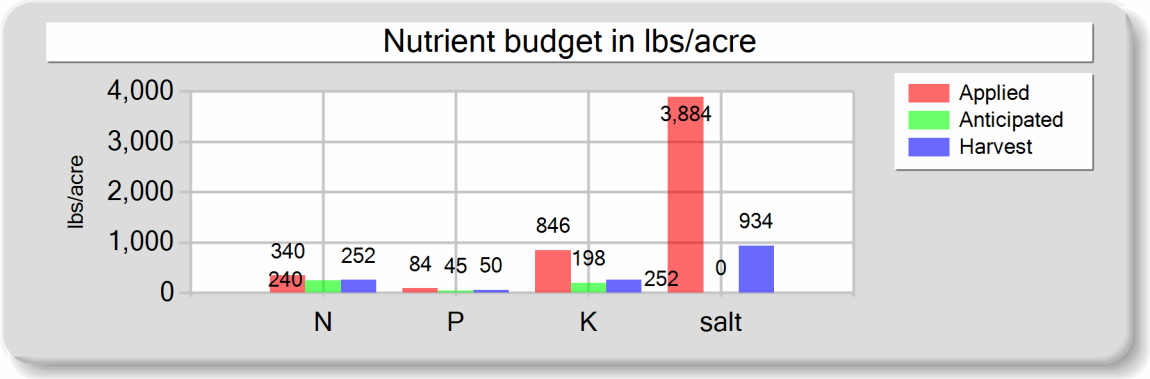
Fresh water applied	
13,368,975.00	*gallons*
492.33	*acre-inches*
12.31	*inches/acre*

Process wastewater applied	
1,560,000.00	*gallons*
57.45	*acre-inches*
1.44	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jersey West - 05/30/2023: Corn, silage

Field name: Jersey West | Crop: Corn, silage | Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	332.84	83.88	846.33	3,505.15
Fresh water	0.00	0.00	0.00	378.81
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	339.84	83.88	846.33	3,883.96
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	251.65	49.57	251.65	934.17
Nutrient balance	88.19	34.32	594.68	2,949.79
Applied to removed ratio	1.35	1.69	3.36	4.16

Fresh water applied	
45,393,900.00	*gallons*
1,671.70	*acre-inches*
41.79	*inches/acre*

Process wastewater applied	
3,590,000.00	*gallons*
132.21	*acre-inches*
3.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure

Sample date: 06/05/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 11.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,600.00	4,200.00	10,200.00							
DL	100.00	100.00	100.00							

Solid Manure

Sample and source description: Solid Manure

Sample date: 10/16/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 4.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,900.00	6,900.00	21,300.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,390.00	332.00	0.00	2.00	0.33	8.16								8,720.00	4,400
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Waste water

Sample and source description: Waste water

Sample date: 06/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	978.00	383.00	0.00	1.00	270.00	1,080.00								9,370.00	2,200
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	444.00	294.00	0.00	0.40	112.00	1,130.00								8,240.00	4,680
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	486.00	347.00	0.00	0.00	89.50	921.00								6,850.00	5,570
DL	1.00	0.50	0.50	0.50	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - DW #1

Domestic Well - DW #1

DW #1

Sample description: DW #1

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			30.80								1,650.00	
DL			0.10								10.00	

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 07/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00										32.80	40
DL	1.00										10.00	10

D. SOIL ANALYSES

Jersey North

Jersey North

Sample and source description: Jersey North

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			197.00				
DL			2.00				

Jersey South

Jersey South

Sample and source description: Jersey South

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			184.00				
DL			2.00				

Jersey West

Jersey West

Sample and source description: Jersey West

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			114.00				
DL			2.00				

E. PLANT TISSUE ANALYSES

Jersey North - 11/19/2022: Oats, silage-soft dough

Jersey North

Sample and source description: Jersey North

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 83.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,500.00	4,300.00	41,700.00		14.10
DL	100.00	100.00	100.00		0.01

Jersey North - 05/30/2023: Corn, silage

Jersey North - 05/30/2023: Corn, silage

Jersey North

Sample and source description: Jersey North

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,500.00	2,500.00	14,000.00		4.70
DL	100.00	100.00	100.00		0.01

Jersey South - 11/19/2022: Oats, silage-soft dough

Jersey South

Sample and source description: Jersey South

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 84.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,200.00	4,800.00	40,700.00		13.70
DL	100.00	100.00	100.00		0.01

Jersey South - 05/30/2023: Corn, silage

Jersey South

Sample and source description: Jersey South

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,600.00	2,500.00	13,100.00		4.30
DL	100.00	100.00	100.00		0.01

Jersey West - 11/19/2022: Oats, silage-soft dough

Jersey West

Sample and source description: Jersey West

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 84.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,500.00	4,600.00	37,000.00		13.70
DL	100.00	100.00	100.00		0.01

Jersey West - 05/30/2023: Corn, silage

Jersey West

Sample and source description: Jersey West

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	2,600.00	13,200.00		4.90
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

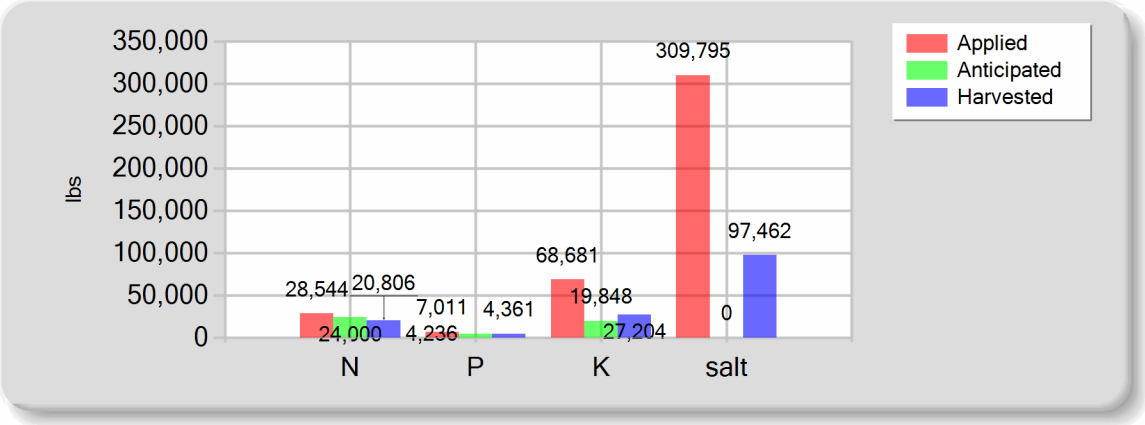
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

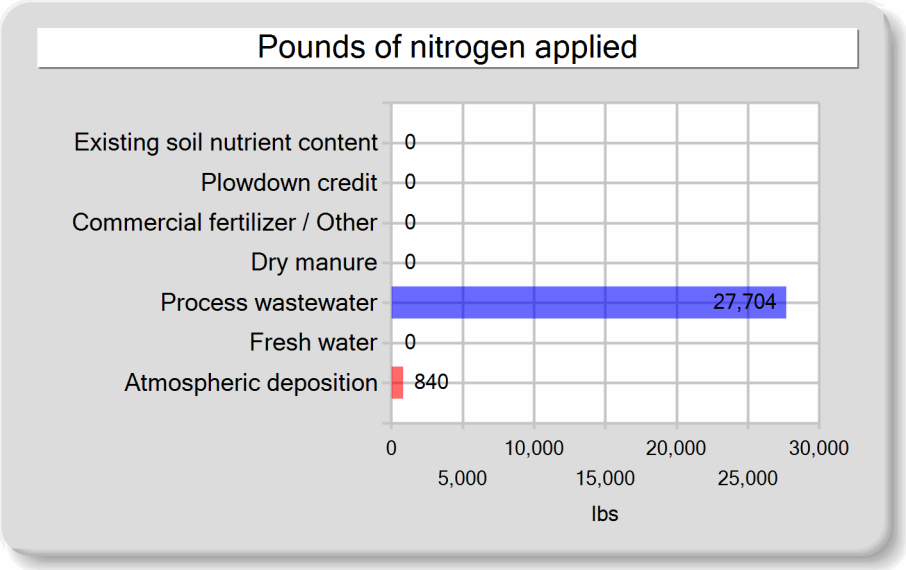
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

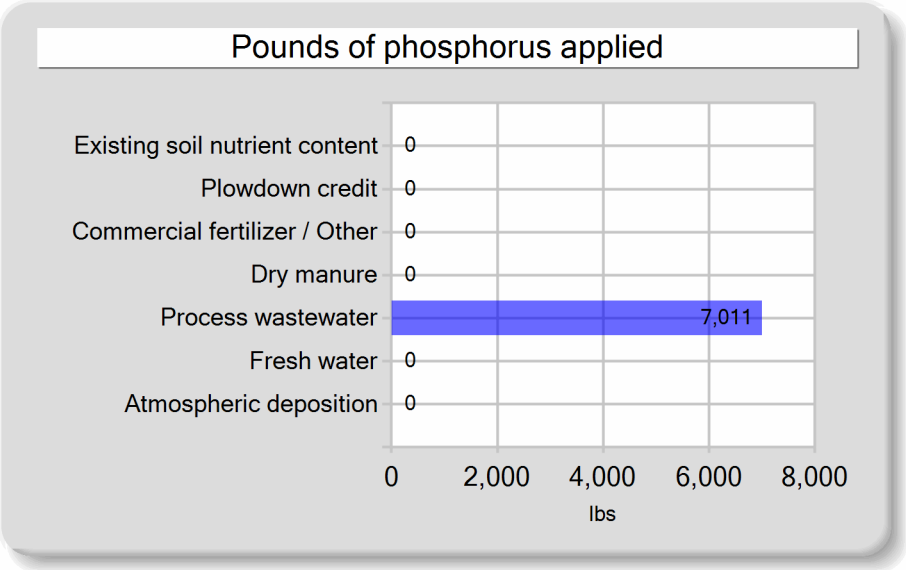
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	27,704.28	7,011.30	68,681.02	281,603.69
Fresh water	0.00	0.00	0.00	28,191.71
Atmospheric deposition	840.00	0.00	0.00	0.00
Total nutrients applied	28,544.28	7,011.30	68,681.02	309,795.41
Anticipated crop nutrient removal	24,000.00	4,236.00	19,848.00	0.00
Actual crop nutrient removal	20,805.70	4,361.42	27,204.33	97,462.43
Nutrient balance	7,738.58	2,649.88	41,476.69	212,332.98
Applied to removed ratio	1.37	1.61	2.52	3.18

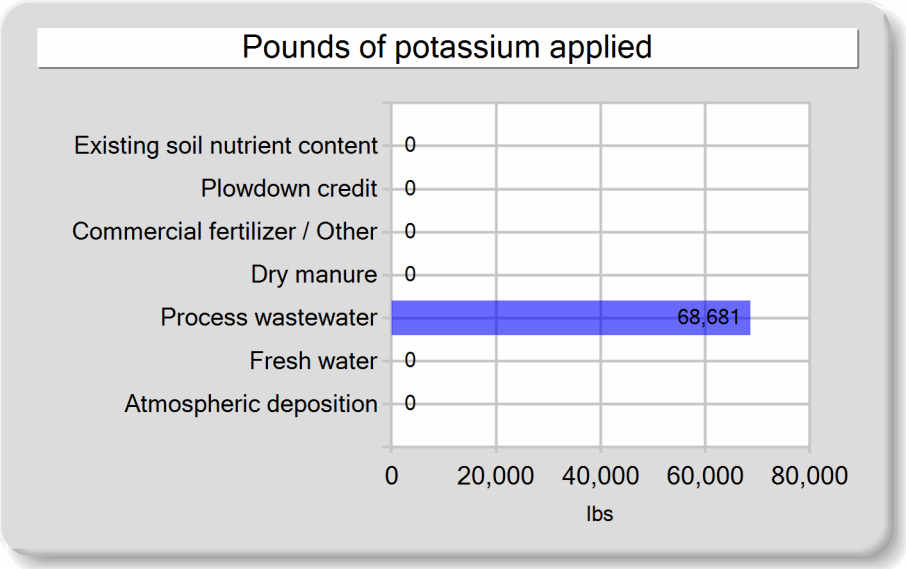
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

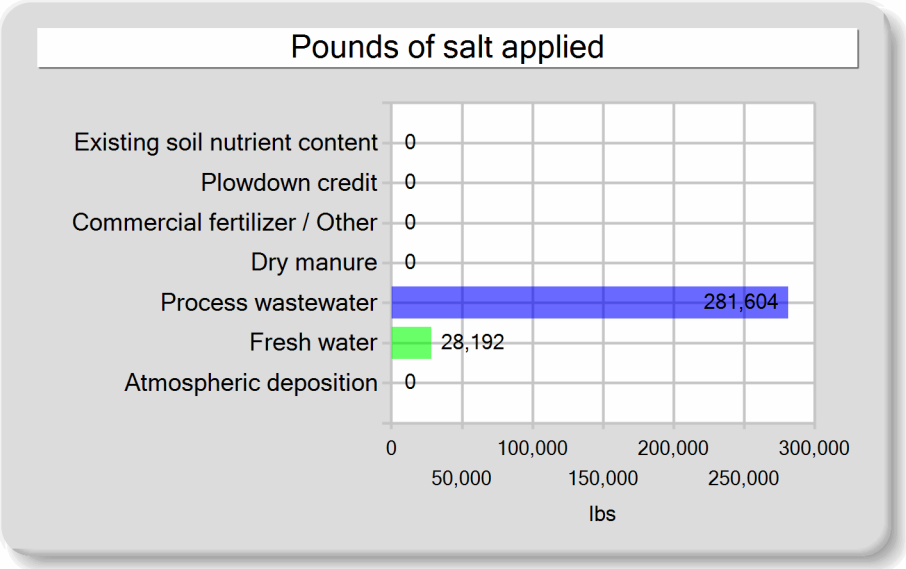


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Arlon Waterson	John Brasil
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/28/2024	6/28/2024
DATE	DATE

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: John Brasil Dairy #2

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:43

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0685-01	DW #1	Ag Water	T. Madruga		11/13/2023 16:00

Default Cooler Temperature on Receipt °C: 4.4
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Sample Results

Sample: DW #1
23K0685-01 (Water)

Sampled: 11/13/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.65**	mmhos/cm	0.01	1		11/17/23 11:44	SM 2510 B		BEK0478
Electrical Conductivity umhos	**1650**	umhos/cm	10.0	1		11/17/23 11:44	SM 2510 B		BEK0478
Ammonia (as N)	ND	mg/L	0.00	1		11/13/23 16:00	Field		BEK0432
Nitrate Nitrogen as NO3N	**30.8**	mg/L	0.1	1	10	11/15/23 13:45	EPA 300.0		BEK0487
Temperature	**25.0**	units	0.0	1		11/17/23 11:44	SM 4500-H+	H	BEK0478
pH	**7.7**	units	1.0	1		11/17/23 11:44	SM 4500-H+	H	BEK0478

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Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0478										
Blank (BEK0478-BLK1)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.0		1.0	units						
Blank (BEK0478-BLK2)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.1		1.0	units						
Blank (BEK0478-BLK3)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.5		1.0	units						
Duplicate (BEK0478-DUP1)		**Source: 23K0682-02**			Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	2.44		0.01	mmhos/cm		2.44			0.0205	10
pH	7.8		1.0	units		7.8			0.128	10
Electrical Conductivity umhos	2440		10.0	umhos/cm		2440			0.0205	10
Duplicate (BEK0478-DUP2)		**Source: 23K0690-02**			Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	1.19		0.01	mmhos/cm		1.18			0.304	10
Electrical Conductivity umhos	1190		10.0	umhos/cm		1180			0.304	10
pH	7.8		1.0	units		7.8			0.128	10
Reference (BEK0478-SRM1)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	562			umhos/cm	538.0		105	90-110		
Reference (BEK0478-SRM2)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEK0478-SRM3)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEK0478-SRM4)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEK0478-SRM5)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	1080			umhos/cm	1000		108	90-110		

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Received: 11/14/2023 15:45
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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0478 (Continued)										
Reference (BEK0478-SRM5)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity umhos	1080			umhos/cm	1000		108	90-110		
Reference (BEK0478-SRM6)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0478-SRM7)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0478-SRM8)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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Ranch: John Brasil Dairy #2

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:43

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0487										
Blank (BEK0487-BLK1)					Prepared & Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0487-BLK2)					Prepared & Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0487-BLK3)					Prepared & Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0487-BS1)					Prepared & Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		97.4	90-110		
LCS (BEK0487-BS2)					Prepared & Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		97.9	90-110		
Duplicate (BEK0487-DUP1)		**Source: 23K0686-01**			Prepared & Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	21.9		0.1	mg/L		21.7			0.720	10
Duplicate (BEK0487-DUP2)		**Source: 23K0720-01**			Prepared & Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	0.05		0.1	mg/L		0.05			7.69	10
Matrix Spike (BEK0487-MS1)		**Source: 23K0686-01**			Prepared & Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	26.8		0.1	mg/L	5.000	21.7	101	90-110		
Matrix Spike (BEK0487-MS2)		**Source: 23K0720-01**			Prepared & Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000	0.05	97.0	90-110		
Reference (BEK0487-SRM1)					Prepared & Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	9.7			mg/L	10.00		96.9	90-110		
Reference (BEK0487-SRM2)					Prepared & Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	9.6			mg/L	10.00		96.2	90-110		
Reference (BEK0487-SRM3)					Prepared: 11/15/2023 Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	9.7			mg/L	10.00		96.6	90-110		

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11/14/23 15:45 23K0685

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

JG

No. Samples	1	No. of Bottles:	1

Purchase Order No	Bill To:	Acct # 19840	Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State: CA** **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT John Brasil Dairy #2
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	DW #1	11/13/2023	1600	4.4	ND
2					
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/13/2023 @ 1600	11/14/2023 @ 900
Second	JLF	Dellavalle Laboratory, Inc.	11/14/2023 @ 900	11/14/23 3:30
Third	[illegible]			
Fourth		DVI	11/14 15:45	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/14/23 15:45 23K0685

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1									
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										